THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Gametes carry 1

(a) half the number of chromosomes **(b)** no chromosomes
(c) only X **(d)** twice the number

2. In pea plants, the recessive form of seed shape is 1

(a) yellow **(b)** green **(c)** round **(d)** wrinkled

3. The zygote formed from X sperm and X egg develops into 1

(a) no baby **(b)** a twin boy **(c)** a girl **(d)** a boy

4. Fertilisation restores the 1

(a) half the number **(b)** zero
(c) double number **(d)** full number of chromosomes

5. A pair of contrasting characters is 1

(a) tall and dwarf **(b)** tall and round
(c) yellow and wrinkled **(d)** green and tall

6. The chance that a newborn is a girl is 1

(a) 25% **(b)** 100% **(c)** 50% **(d)** 75%

7. A bodybuilder's well-developed muscles will be passed to children? 1

(a) Yes **(b)** Only to sons
(c) Only to daughters **(d)** No

8. Mendel's first generation of the cross is called 1

(a) P **(b)** F3 **(c)** F2 **(d)** F1

9. Who decides the sex of the child? 1

(a) Doctor **(b)** Father **(c)** Grandmother **(d)** Mother

10. Alleles are 1

(a) different forms of a gene **(b)** gametes
(c) two chromosomes **(d)** proteins

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Give one example of an acquired trait. 2

12. Name the plant on which Mendel experimented. 2

13. What does F1 stand for? 2

14. Which parent determines the sex of the child? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the genotype of a pure tall and a pure dwarf pea plant. State the genotype of F1. 3

16. How are the chromosomes passed from one generation to the next? 3

17. Name the contrasting traits of seed colour and seed shape in pea. 3

OR

State the proportion of tall and dwarf plants in the F2 generation in a monohybrid cross. 3

18. Why do the children in a family look different from each other? 3

OR

Name the type of cross between parents differing in two traits and give the F2 ratio. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how variation arises in sexual reproduction. 5

OR

Describe how the number of chromosomes remains the same in successive generations. 5

20. Why is the garden pea suitable for Mendel's experiments? Give four reasons. 5

OR

Why do some traits not appear in the F1 generation but appear in the F2? Explain. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND